

Unit 4 Review*Solving Problems with Rational Numbers*

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Adding and Subtracting in Scientific Notation*Lesson 1 · 6 questions*

1. $(4.1 \times 10^6) + (2.3 \times 10^6)$

2. $(7.6 \times 10^{-4}) - (2.9 \times 10^{-4})$

3. $(8.2 \times 10^5) + (4.9 \times 10^5)$

4. $(5.4 \times 10^8) + (7.1 \times 10^7)$

5. $(6.3 \times 10^{-2}) - (8.0 \times 10^{-4})$

6. Lena writes $(3.2 \times 10^4) + (5.5 \times 10^6) = 8.7 \times 10^{10}$. Name her two mistakes and give the correct sum.

PART B. Multiplying and Dividing in Scientific Notation

Lessons 2–3 · 6 questions

7. $(4 \times 10^3)(2 \times 10^6)$

8. $\frac{7.2 \times 10^8}{1.8 \times 10^3}$

9. $(5 \times 10^6)(8 \times 10^{-2})$

10. $\frac{2.4 \times 10^5}{8 \times 10^9}$

11. $\frac{4 \times 10^7}{4 \times 10^{-3}}$

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4.0×10^2 seconds?

PART C. Real-World Problems and Significant Digits

Lesson 4 · 6 questions

13. How many significant digits are in each measurement? 6,040 g · 0.00920 m · 45.00 s
14. A rectangular panel measures 3.2×10^2 mm by 1.45×10^2 mm. Find its area, with the correct number of significant digits.
15. A train travels 4.80×10^2 km in 6.0 hours. Find its average speed, with the correct number of significant digits.
16. Two lengths are measured: 5.40×10^3 mm and 2.8×10^2 mm. Find their sum, with the correct precision.

17. Two masses are measured: 7.60×10^{-2} g and 3.45×10^{-3} g. Find their difference, with the correct precision.
18. Micah adds two measured lengths, 2,340 m and 6.8 m, and says the sum must be written with two significant digits, as 2.3×10^3 m. Is he right? Explain, and give the correct sum.

PART D. Evaluating Expressions with Radicals

Lesson 5 · 7 questions

19. $\sqrt{169}$

20. $\sqrt[3]{-8}$

21. $\sqrt{7^2 - 24}$

22. $\sqrt[3]{\frac{1}{27}}$

23. $\sqrt{100} \cdot 4^{-2}$

24. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2 + 7}$

25. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

PART E. The Order of Operations

Lesson 6 · 7 questions

26. $24 - 3 \cdot 5 + 1$

27. $48 \div 6 \cdot 2$

28. $4 + 3(8 - 5)^2$

29. $\frac{6^2+9}{3^2-4}$

30. $-6^2 + (-6)^2$

31. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

32. Which expression is equivalent in value to $2 + 4 \cdot 5^2$?

- ☐ A. 102
- ☐ B. 150
- ☐ C. 900
- ☐ D. 42

PART F. Real-World Problems and the Order of Operations

Lessons 7–8 · 5 questions

33. Evaluate $5x^2$ when $x = -3$.

34. Kinetic energy is $K = \frac{1}{2}mv^2$. Find K when $m = 400$ kg and $v = 7$ m/s.

35. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

36. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery.

a. What is the area of the patio?

b. Write an expression for the total cost.

c. Find the total cost.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

37. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ☐ A. $18^2(3.50) + 140$
- ☐ B. $18^2(3.50 + 140)$
- ☐ C. $18(3.50) + 140$
- ☐ D. $18^2 + 3.50 + 140$